CS3000 Algorithms & Data **Practice Quiz 5A**	**Name:** _____ **NUID:** _____

Instructions:

- Please put your phones on silent and in your bag for the duration of the quiz.
- This quiz is designed to last 40 minutes.
- If you have a question, alert a proctor quietly, and we'll be glad to assist.
- If we can't read your answer, we can't grade your answer. Please write legibly.
- Practice version: the format and length match the real quizzes, but these are new problems.

Q1. A vending machine gives change using coins of denominations $d_1 = 1$, $d_2 = 4$, and $d_3 = 6$ (unlimited supply of each). Let $C(v)$ be the minimum number of coins needed to make exactly the amount v.

(a) Write the recurrence for $C(v)$, including the base case.

(b) Fill in the table bottom-up for $v = 0, 1, \ldots, 9$.

v	0	1	2	3	4	5	6	7	8	9
$C(v)$										

(c) The greedy strategy repeatedly takes the largest coin that fits. What does greedy return for $v = 9$, and what is the optimal answer? Which coins does the optimal answer use?

(d) Modify the bottom-up `COIN-CHANGE` pseudocode so that it also returns the list of coins used. State its running time and extra space in terms of v and the number of denominations k.

Q1 (continued).

Q2. Let $X =$ `TACGA` and $Y =$ `ACTGAC`. Let $L(i, j)$ be the length of the longest common subsequence of the first i characters of X and the first j characters of Y.

(a) Fill in the table bottom-up using the recurrence from class.

$X_i \downarrow\ Y_j \rightarrow$	0	1 (A)	2 (C)	3 (T)	4 (G)	5 (A)	6 (C)
0							
1 (T)							
2 (A)							
3 (C)							
4 (G)							
5 (A)							

(b) What is the length of the LCS? Starting from $L(5, 6)$, trace back through the table to recover an LCS. Circle the cells on your path.

(c) Write pseudocode for `LCS-TRACEBACK(L, X, Y)` that returns the LCS string after the table L has been filled. What is its running time?

Q2 (continued).